

FWMamba-UNet: Frequency-Wavelet Enhanced Mamba UNet for Medical Image Segmentation

Zihao Guo^{1*}(✉), Haoyu Yang^{1*}, Zehan Zhao^{1*}, Jiaming Xu^{1*}, Hongyan Yin¹, and Yuechen Zhang¹

¹ Jiangnan University, Wuxi 214122, China
zihao3351@gmail.com

Abstract. State Space Models (SSMs) such as Mamba have emerged as an efficient backbone for visual recognition, providing global context modeling at linear cost. Yet on dense prediction tasks like medical image segmentation, their outputs consistently show blurred and eroded boundaries. We trace this to a property we term Structural High-Frequency Attenuation (SHFA): under HiPPO initialization, an SSM’s transfer function has vanishing variation beyond an eigenvalue-determined cutoff, and a gradient decay of $\mathcal{O}(|s|^{-2})$ keeps training in the low-frequency regime. To address SHFA without altering the SSM’s internal dynamics, we propose FWMamba-UNet, which routes high-frequency information through an external parallel pathway via (1) the FW-Mamba Block, a Discrete Wavelet Transform (DWT) branch with adaptive band weighting, and (2) the EAFF-Skip connection, which converts directional wavelet subbands into edge-aware attention. With only 10.1% additional parameters, FWMamba-UNet attains state-of-the-art performance on Synapse, ACDC, and ISIC 2018. Ablations further show that SHFA is a feature-representation deficiency that must be addressed architecturally rather than via loss supervision. Our code is available at <https://github.com/unumbrela/FWMamba-UNet>.

Keywords: Medical Image Segmentation, Mamba, Spectral Bias, Wavelet Transform.

1 Introduction

Accurate medical image segmentation underpins computer-aided diagnosis, treatment planning, and disease monitoring [11]. As a dense prediction problem in computer vision, it benefits from both local detail and global context. CNN-based U-Nets [20][25] capture local features but have limited receptive fields; Transformers [2][4] enlarge them via self-attention at quadratic cost. State Space Models (SSMs), and Mamba [7] in particular, combine linear complexity with strong global modeling, motivating a recent wave of Mamba-based segmentation architectures such as VM-UNet [21], U-Mamba [15], and Swin-UMamba [13].

Beneath this progress lies a structural limitation of SSMs that has received little attention in the vision community, which we term *Structural High-Frequency Attenuation* (SHFA). Building on Yu et al. [24], HiPPO-initialized SSMs [8] are intrinsically

Z. Guo, H. Yang, J. Xu, Z. Zhao—Equal contribution.

biased toward low frequencies: their transfer function has vanishing variation beyond an eigenvalue-determined cutoff and gradient sensitivity decays as $O(|s|^{-2})$, trapping the dynamics in a low-frequency regime. **Fig. 1** illustrates this on a real dermoscopic image: high-frequency energy encoding lesion boundaries is almost entirely lost, while low-frequency structure is preserved. InceptionMamba [9] reports the same bias on classification, indicating that SHFA is a general property of Mamba-style vision backbones.

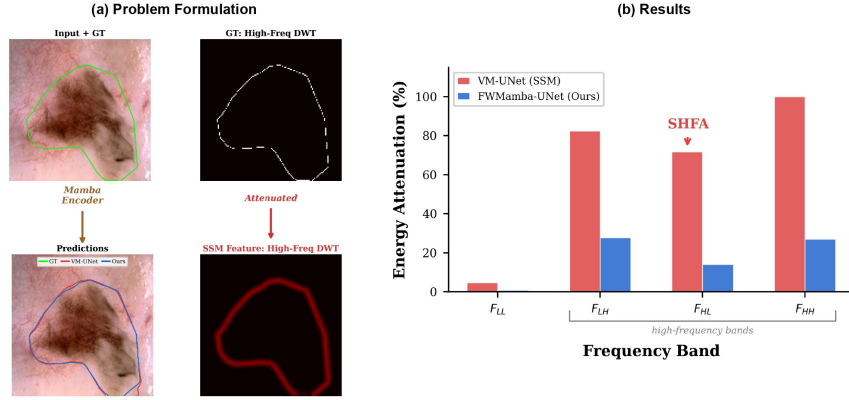


Fig. 1. Structural High-Frequency Attenuation (SHFA) of SSMs on a real ISIC 2018 dermoscopic image. (a) The GT segmentation boundary resides in DWT high-frequency subbands (top right, bright edges). The SSM’s feature-level representation severely attenuates this energy (bottom right). Prediction contours are from trained VM-UNet (red) and FWMamba-UNet (blue). (b) Per-band energy attenuation: the SSM loses up to 90% of high-frequency energy (F_{LH} – F_{HH}) while preserving low-frequency structure (F_{LL} , <5%). Our DWT branch reduces attenuation to ~20%.

Existing Mamba segmentation methods explore new scanning strategies [14], CNN-Mamba hybridization [15], or richer local interactions [23], but none is explicitly designed to compensate for SHFA. A second line introduces spectral signals into Mamba: CFG-MambaNet [19] attaches a frequency module outside the SSM pathway, and WMC-Net [6] injects wavelets into the S6 dynamics. Both confirm that frequency cues help, yet neither is derived from a transfer-function-level account of *why* SSMs underperform on high-frequency content.

Motivated by this gap, we propose **FWMamba-UNet**: instead of altering the SSM’s internal dynamics, we keep its low-frequency pathway intact and route high-frequency information through an *external parallel branch*—exactly the component the SSM provably cannot learn on its own. Our contributions are:

1. We identify SHFA as a structural cause of boundary degradation in SSM-based dense prediction and propose the **FW-Mamba Block**, a parallel DWT branch with adaptive band weighting.
2. We design **EAFF-Skip**, which turns directional DWT subbands into edge-aware attention for boundary-preserving feature propagation.

3. Experiments on three benchmarks set new state-of-the-art with only +10.1% parameters; ablations show SHFA is primarily a *feature-extraction* deficiency, yielding a transferable design principle for SSM-based vision.

2 Related Work

Mamba-Based Medical Image Segmentation. VM-UNet [21] establishes a VSS-based U-shaped framework, while U-Mamba [15], Mamba-UNet [22], Swin-UMamba [13], and H-vmunet [23] further explore CNN-Mamba hybridization, pure or pretrained Mamba backbones, and high-order interactions. These methods center on scanning patterns, hybridization, or interaction design rather than the SSM’s intrinsic spectral behavior.

Spectral Bias of SSMs and Frequency Enhancement. Frequency bias of neural networks was first observed in MLPs [18]. For SSMs, Yu et al. [24] provide the first rigorous characterization, linking transfer-function variation to eigenvalue distribution under HiPPO initialization. CFG-MambaNet [19] and WMC-Net [6] introduce FFT- or wavelet-based enhancement, yet are not explicitly grounded in this analysis. We instead preserve the original SSM pathway and add a parallel wavelet branch as a principled high-frequency compensation.

3 Methodology

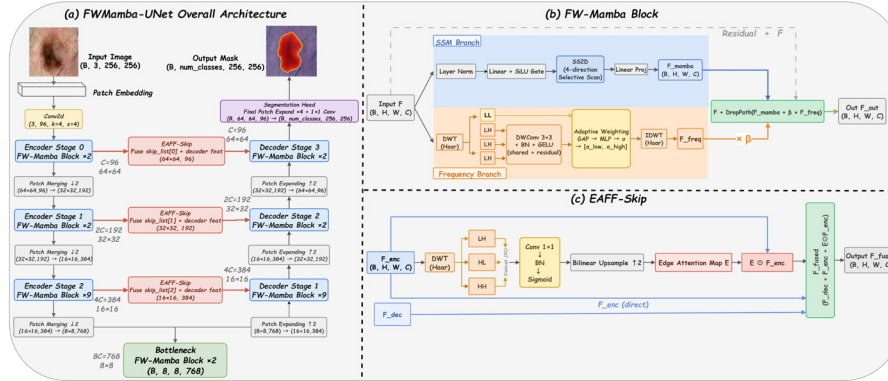


Fig. 2. Overview of FWMamba-UNet. (a) U-shaped architecture with FW-Mamba Blocks and EAFF-Skip connections. (b) FW-Mamba Block: the SSM branch captures global dependencies while a parallel DWT branch enhances high-frequency components with adaptive band weighting. (c) EAFF-Skip: directional high-frequency subbands generate edge attention maps for boundary-enhanced skip connections.

3.1 Preliminaries and Theoretical Motivation

State Space Models. A continuous-time SSM maps input $x(t)$ to output $y(t)$ via $h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t)$, $y(t) = \mathbf{C}h(t)$. After discretization with step size Δ , this becomes $h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t$, $y_t = \mathbf{C}h_t$. In the Fourier domain, the system acts as a filter via its transfer function $G(is) = \mathbf{C}(is\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$ [24]. Mamba [7] introduces input-dependent parameters for content-aware modeling with linear complexity; SS2D [14] extends this to 2D via four-directional scanning.

Structural High-Frequency Attenuation (SHFA). Consider a diagonal matrix $\mathbf{A} = \text{diag}(a_1, \dots, a_n)$ with $a_j = x_j + iy_j$ and $x_j < 0$. Its real-valued transfer function admits the following partial fraction decomposition [24]:

$$\tilde{G}(is) = \sum_{j=1}^n \frac{\zeta_j(s-y_j) - \xi_j x_j}{x_j^2 + (s-y_j)^2} + D, c_j = \xi_j + i\zeta_j \quad (1)$$

where each fraction resonates near $s = y_j$. The system’s ability to distinguish frequencies is measured by total variation $V_a^b(\tilde{G})$. For $B > \max_j |y_j|$, Yu et al. [24] prove:

$$V_B^\infty(\tilde{G}) \leq \sum_{j=1}^n \frac{|c_j|}{|y_j - B|} \quad (2)$$

Under HiPPO initialization where $|y_j| = \mathcal{O}(n)$, this bound vanishes for $B = \omega(n)$ —the SSM structurally cannot distinguish high-frequency content, hence the term SHFA. This bias persists through training: the gradient kernel $K_j(s)$ decays as $\mathcal{O}(|s|^{-2})$, trapping eigenvalues in low-frequency minima changes only 1.43% over 100 epochs [24].

Since organ boundaries and lesion edges reside in high-frequency subbands that the SSM provably attenuates, our design principle follows naturally: rather than modifying the SSM’s internal dynamics, we introduce an external parallel pathway that supplies the high-frequency representation it cannot capture.

Discrete Wavelet Transform. The 2D DWT decomposes $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$ into subbands $\{\mathbf{F}_{LL}, \mathbf{F}_{LH}, \mathbf{F}_{HL}, \mathbf{F}_{HH}\}$, where $\mathbf{F}_{LL}$ captures low-frequency approximation and $\mathbf{F}_{LH}$, $\mathbf{F}_{HL}$, $\mathbf{F}_{HH}$ capture directional edge details (cf. Fig. 1 1a, top right). We use fixed Haar wavelets [16]. DWT is preferred over FFT for its *spatially localized* decomposition—preserving spatial correspondence between sub-bands and boundary locations, a critical property for pixel-level segmentation.

3.2 FW-Mamba Block

The FW-Mamba Block (Fig. 2) replaces the standard VSS block with a dual-branch design: the SSM branch for global context, and a parallel DWT branch for high-frequency compensation.

Frequency Enhancement Branch. Given $\mathbf{F} \in \mathbb{R}^{B \times C \times H \times W}$, we obtain subbands via DWT and refine high-frequency components:

$$\hat{\mathbf{F}}_s = \mathbf{F}_s + \text{DWConv}_{3 \times 3}(\mathbf{F}_s), s \in \{LH, HL, HH\}, \quad (3)$$

where $\text{DWConv}_{3 \times 3}$ denotes depthwise convolution with BatchNorm and GELU. The three subbands share weights, adding only $9C$ parameters per block.

Adaptive Frequency Band Weighting. Shallow layers (richer boundary content) need stronger compensation than deeper layers (semantic abstractions). Learnable weights enable stage-adaptive control:

$$[\alpha_{\text{low}}, \alpha_{\text{high}}] = \sigma(\text{MLP}(\text{GAP}(\mathbf{F}))), \quad (4)$$

where MLP has bottleneck ratio 8 ($C \rightarrow C/8 \rightarrow 2$). The enhanced feature is reconstructed via IDWT:

$$\mathbf{F}_{\text{freq}} = \text{IDWT} \left(\alpha_{\text{low}} \cdot \mathbf{F}_{LL}, \alpha_{\text{high}} \cdot \hat{\mathbf{F}}_{LH}, \alpha_{\text{high}} \cdot \hat{\mathbf{F}}_{HL}, \alpha_{\text{high}} \cdot \hat{\mathbf{F}}_{HH} \right). \quad (5)$$

Dual-Branch Fusion. The branches are fused with learnable scalar β (initialized to 0.1 for training stability):

$$\mathbf{F}_{\text{out}} = \mathbf{F} + \text{DropPath} \left(\underbrace{\text{VSSBlock}(\mathbf{F})}_{\text{global context}} + \beta \cdot \underbrace{\mathbf{F}_{\text{freq}}}_{\text{high-freq compensation}} \right). \quad (6)$$

3.3 Edge-Aware Frequency Fusion Skip Connection

In Mamba architectures, encoder features forwarded via skip connections are *already deficient in high-frequency boundary information* due to SHFA. EAFF-Skip recovers boundary signals by extracting high-frequency components directly via DWT. Given encoder features $\mathbf{F}_{\text{enc}}$, we generate a spatial edge attention map:

$$\mathbf{E} = \sigma(\text{Conv}_{1 \times 1}([\mathbf{F}_{LH}; \mathbf{F}_{HL}; \mathbf{F}_{HH}])), \quad (7)$$

The fused output preserves all encoder information while selectively amplifying boundaries:

$$\mathbf{F}_{\text{fused}} = \mathbf{F}_{\text{dec}} + \mathbf{F}_{\text{enc}} + \mathbf{E} \odot \mathbf{F}_{\text{enc}}. \quad (8)$$

3.4 Architecture and Loss Function

FWMamba-UNet adopts VM-UNet’s [21] symmetric U-shape, replacing VSS blocks with FW-Mamba Blocks and skip connections with EAFF-Skip modules. The encoder

has four stages with dimensions $[C, 2C, 4C, 8C]$ ($C = 96$), each containing two FWMamba Blocks. SSM components are initialized from pretrained VMamba-Small weights [14]. Total: 30.31M parameters (baseline 27.53M + 2.78M frequency modules, +10.1%).

The composite loss combines standard objectives with lightweight frequency-aware regularization:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}} + \lambda_e \cdot \mathcal{L}_{\text{edge}} + \lambda_f \cdot \mathcal{L}_{\text{freq}}, \quad (9)$$

where $\mathcal{L}_{\text{edge}}$ is a Sobel-weighted Dice loss emphasizing boundaries ($w_i = 1 + \gamma \|\nabla_{\text{Sobel}} g\|_i$, $\gamma = 2$), and $\mathcal{L}_{\text{freq}}$ penalizes high-frequency discrepancies in Fourier space. The weights $\lambda_e = 0.1$, $\lambda_f = 0.02$ are deliberately small—our ablation confirms that SHFA must be addressed architecturally, not through loss alone (Section 4.3).

4 Experiments

4.1 Setup

Datasets. (1) *Synapse* [12]: 30 abdominal CT cases, 8 organs, 18/12 train/test split, 224×224 . (2) *ACDC* [1]: 100 cardiac MRI cases, 3 structures (RV, Myo, LV), 70/10/20 split. (3) *ISIC* 2018 [5]: 2,594 dermoscopic images, binary segmentation, 70/30 train/val split, 256×256 .

Implementation. AdamW optimizer, lr 10^{-3} , weight decay 10^{-2} , cosine annealing, 300 epochs. Batch size 16. VMamba-Small pretrained weights. Metrics: DSC (%), HD95 (mm) for Synapse/ACDC; DSC, mIoU, Sensitivity for ISIC.

4.2 Comparison with State-of-the-Art

Synapse. Over three repeated runs, FWMamba-UNet achieves $83.45 \pm 0.24\%$ DSC and 17.80 ± 0.30 mm HD95, outperforming VM-UNet by +4.89% DSC and 7.79 mm HD95 in mean performance, and establishing new state-of-the-art on both metrics (**Table 1**). Gains are most pronounced on organs with complex boundaries—gallbladder (+8.95%) and pancreas (+7.25%)—consistent with SHFA: these structures rely on high-frequency edge cues that the SSM’s transfer function severely attenuates. It also surpasses U-Mamba by 1.39% DSC and WMC-Net by 2.89%, suggesting that principled spectral compensation is more effective than ad hoc frequency module insertion.

ACDC. Over three repeated runs, FWMamba-UNet achieves $93.28 \pm 0.18\%$ DSC, surpassing CFG-MambaNet by +0.54% in mean performance and establishing a new state of the art (**Table 2**). The largest gain is on the myocardium (+5.42% over VM-UNet), the thinnest structure and thus most sensitive to SHFA—directly supporting our theoretical analysis.

ISIC 2018. Over three repeated runs, FWMamba-UNet achieves $89.57 \pm 0.19\%$ DSC and $91.26 \pm 0.28\%$ Sensitivity (+0.74% DSC and +1.53% Sensitivity over VM-UNet, **Table 3**). The disproportionate Sensitivity gain reflects that skin lesion boundaries involve gradual color transitions concentrated in the mid-to-high frequency range most severely attenuated by SHFA. Consistent improvements across CT, MRI, and dermoscopy further demonstrate that our approach targets a fundamental SSM property rather than a modality-specific artifact.

Table 1. Comparison on Synapse. Publication venue/year is shown after each base-line. Results are reported as mean $\pm$ std over three runs. Best bold, second underlined, ranked by the mean.

Method	DSC $\uparrow$	HD95 $\downarrow$	Aorta	Gall.	L.Kid	R.Kid	Liver	Panc.	Spl.	Stom.
<i>CNN-based</i>										
U-Net (MICCAI'15) [20]	76.85 $\pm$ 0.41	39.70 $\pm$ 0.92	89.07 $\pm$ 0.28	<u>69.72$\pm$0.54</u>	77.77 $\pm$ 0.35	68.60 $\pm$ 0.39	93.43 $\pm$ 0.20	53.98 $\pm$ 0.58	86.67 $\pm$ 0.31	75.58 $\pm$ 0.37
AttU-Net [17]	77.77 $\pm$ 0.38	36.02 $\pm$ 0.84	89.55$\pm$0.26	68.88 $\pm$ 0.49	77.98 $\pm$ 0.33	71.11 $\pm$ 0.36	93.57 $\pm$ 0.19	58.04 $\pm$ 0.53	87.30 $\pm$ 0.29	75.75 $\pm$ 0.35
<i>Transformer-based</i>										
TransUNet (MedIA'24) [4]	77.48 $\pm$ 0.35	31.69 $\pm$ 0.76	87.23 $\pm$ 0.24	63.13 $\pm$ 0.44	81.87 $\pm$ 0.30	77.02 $\pm$ 0.34	94.08 $\pm$ 0.18	55.86 $\pm$ 0.47	85.08 $\pm$ 0.27	75.62 $\pm$ 0.33
Swin-UNet (ECCVW'23) [2]	79.13 $\pm$ 0.31	21.55 $\pm$ 0.58	85.47 $\pm$ 0.22	66.53 $\pm$ 0.37	83.28 $\pm$ 0.27	79.61 $\pm$ 0.29	94.29$\pm$0.16	56.58 $\pm$ 0.41	90.66 $\pm$ 0.23	76.60 $\pm$ 0.28
MISSFormer (TMI'23) [10]	81.96 $\pm$ 0.27	<u>18.20$\pm$0.39</u>	86.99 $\pm$ 0.20	68.65 $\pm$ 0.31	85.21 $\pm$ 0.24	82.00 $\pm$ 0.25	94.41 $\pm$ 0.14	65.67$\pm$0.35	91.92 $\pm$ 0.20	80.81 $\pm$ 0.24
<i>Mamba-based</i>										
VM-UNet (TOMM'25) [21]	78.56 $\pm$ 0.33	25.59 $\pm$ 0.61	85.48 $\pm$ 0.23	62.85 $\pm$ 0.41	82.42 $\pm$ 0.29	79.96 $\pm$ 0.31	93.75 $\pm$ 0.17	56.45 $\pm$ 0.45	89.53 $\pm$ 0.25	78.00 $\pm$ 0.31
Mamba-UNet [22]	79.07 $\pm$ 0.30	23.48 $\pm$ 0.56	85.80 $\pm$ 0.21	63.10 $\pm$ 0.39	82.91 $\pm$ 0.27	80.50 $\pm$ 0.29	94.21 $\pm$ 0.16	57.10 $\pm$ 0.42	89.70 $\pm$ 0.23	78.96 $\pm$ 0.29
Swin-UMamba [13]	79.48 $\pm$ 0.28	22.10 $\pm$ 0.50	86.05 $\pm$ 0.20	64.20 $\pm$ 0.36	83.56 $\pm$ 0.25	80.12 $\pm$ 0.27	94.55 $\pm$ 0.15	57.80 $\pm$ 0.40	90.20 $\pm$ 0.21	79.36 $\pm$ 0.27
U-Mamba [15]	<u>82.06$\pm$0.23</u>	18.83 $\pm$ 0.34	87.25 $\pm$ 0.17	67.47 $\pm$ 0.28	<u>86.14$\pm$0.20</u>	83.91$\pm$0.24	94.43 $\pm$ 0.12	<u>63.91$\pm$0.28</u>	<u>92.53$\pm$0.18</u>	80.83 $\pm$ 0.22
H-vmunet (NC'25) [23]	80.61 $\pm$ 0.26	19.79 $\pm$ 0.37	86.23 $\pm$ 0.18	64.97 $\pm$ 0.31	84.47 $\pm$ 0.22	81.68 $\pm$ 0.24	<u>94.79$\pm$0.13</u>	60.32 $\pm$ 0.32	91.45 $\pm$ 0.19	<u>80.96$\pm$0.22</u>
WMC-Net (KBS'26) [6]	80.56 $\pm$ 0.27	24.76 $\pm$ 0.52	86.45 $\pm$ 0.19	66.25 $\pm$ 0.33	83.85 $\pm$ 0.23	81.22 $\pm$ 0.25	94.58 $\pm$ 0.14	62.48 $\pm$ 0.34	91.15 $\pm$ 0.20	78.50 $\pm$ 0.24
FWMamba-UNet(Ours)	83.45$\pm$0.24	17.80$\pm$0.30	<u>89.12$\pm$0.18</u>	71.80$\pm$0.27	86.50$\pm$0.20	<u>83.15$\pm$0.23</u>	96.60$\pm$0.12	63.70 $\pm$ 0.29	93.20$\pm$0.18	81.93$\pm$0.22

Table 2. Comparison on ACDC. Publication venue/year is shown after each base-line. Results are reported as mean $\pm$ std over three runs. DSC (%). Best bold, second underlined, ranked by the mean.

Method	Avg. DSC $\uparrow$	RV	Myo	LV
<i>CNN / Transformer-based</i>				
U-Net (MICCAI'15) [20]	87.55 $\pm$ 0.24	87.10 $\pm$ 0.29	80.63 $\pm$ 0.41	94.92 $\pm$ 0.18
TransUNet (MedIA'24) [4]	89.71 $\pm$ 0.20	88.86 $\pm$ 0.25	84.54 $\pm$ 0.35	95.73 $\pm$ 0.16
Swin-UNet (ECCVW'23) [2]	90.00 $\pm$ 0.19	88.55 $\pm$ 0.23	85.62 $\pm$ 0.32	95.83 $\pm$ 0.15
MISSFormer (TMI'23) [10]	90.86 $\pm$ 0.17	89.55 $\pm$ 0.21	88.04 $\pm$ 0.29	94.99 $\pm$ 0.14
<i>Mamba-based</i>				
VM-UNet (TOMM'25) [21]	90.19 $\pm$ 0.19	88.76 $\pm$ 0.23	85.90 $\pm$ 0.33	95.91 $\pm$ 0.15
Mamba-UNet (arXiv'24) [22]	89.65 $\pm$ 0.21	87.37 $\pm$ 0.25	85.82 $\pm$ 0.36	95.76 $\pm$ 0.17
Swin-UMamba (MICCAI'24)[13]	90.28 $\pm$ 0.18	88.90 $\pm$ 0.22	86.18 $\pm$ 0.31	95.76 $\pm$ 0.15

U-Mamba (arXiv'24) [15]	90.53±0.17	89.45±0.20	86.30±0.28	95.84±0.13
CFG-MambaNet(npj DM'26)[19]	<u>92.74±0.16</u>	<u>91.55±0.19</u>	<u>90.55±0.26</u>	<u>96.12±0.12</u>
FWMamba-UNet	93.28±0.18	92.12±0.21	91.32±0.27	96.40±0.13

Table 3. Comparison on ISIC 2018. Publication venue/year is shown after each base-line. Results are reported as mean±std over three runs. Best bold, second underlined, ranked by the mean.

Method	DSC↑	mIoU↑	Sensitivity↑
<i>CNN / Transformer-based</i>			
U-Net (MICCAI'15) [20]	87.55±0.22	77.86±0.31	87.42±0.35
AttU-Net (arXiv'18) [17]	87.83±0.21	78.32±0.29	88.15±0.33
TransUNet (MedIA'24) [4]	88.40±0.20	79.28±0.27	88.76±0.30
Swin-UNet (ECCVW'23) [2]	88.15±0.21	78.94±0.28	88.50±0.31
<i>Mamba-based</i>			
VM-UNet (TOMM'25) [21]	88.83±0.18	79.90±0.25	<u>89.73±0.29</u>
Mamba-UNet (arXiv'24) [22]	<u>88.96±0.17</u>	<u>80.12±0.23</u>	89.50±0.27
FWMamba-UNet	89.57±0.19	81.11±0.24	91.26±0.28

Table 4. Ablation on ISIC 2018. FW = FW-Mamba Block, EAFF = EAFF-Skip, FWL = frequency-aware loss.

FW	EAFF	FWL	DSC↑	mIoU↑	Sensitivity↑
			88.13	79.30	89.33
✓			89.10	80.31	89.52
	✓		89.22	80.18	89.95
		✓	88.91	80.05	90.48
✓	✓		89.32	80.70	91.08
✓	✓	✓	89.57	81.11	91.26

4.3 Ablation Study

Component Analysis. Table 4 reveals a clear pattern consistent with our SHFA analysis. The architectural modules (FW: +0.97% DSC; EAFF: +1.09%; combined: +1.19%) directly compensate for the attenuated high-frequency representation. Combining FW and EAFF also boosts Sensitivity to 91.08%, confirming that parallel frequency compensation and boundary-aware skip connections are complementary. In contrast, FWL alone yields only +0.78% DSC despite +1.15% Sensitivity—loss supervision can encourage boundary-accurate predictions but cannot overcome the SSM’s structural inability to represent high-frequency features (Eq. 2). This establishes a key principle: SHFA is a feature extraction problem, not a supervision problem.

Fig. 3 provides visual confirmation. The +FWL variant shows attention nearly identical to the Baseline, whereas +FW sharply concentrates activation around lesion boundaries. Loss-level supervision alone cannot restructure internal representations; the bias is structurally embedded in the transfer function.

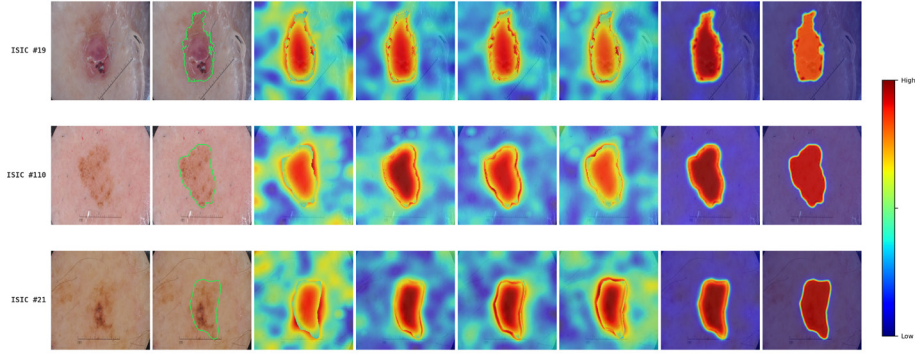


Fig. 3. Grad-CAM visualization on ISIC 2018. Baseline shows diffuse attention; +FW tightens activation toward the lesion; +EAF enhances boundary response; +FWL remains similar to Baseline (loss cannot restructure representations). The full model achieves precise, boundary aware activation.

Loss Weight Sensitivity. Aggressive weights $(\lambda_e, \lambda_f) = (0.5, 0.1)$ degrade DSC to 88.56% (-0.76% vs. no-FWL), while lightweight $(0.1, 0.02)$ yields the best results (89.57% DSC). **Fig. 4** confirms this: overly strong frequency regularization causes persistent optimization conflict, whereas lightweight supervision stabilizes late-stage convergence—reinforcing that frequency-aware loss is effective only as a weak auxiliary signal.

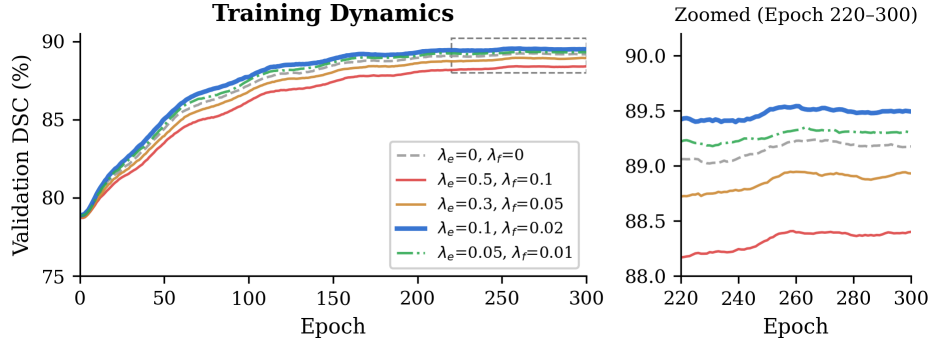


Fig. 4. Validation DSC curves on ISIC 2018 for different frequency-aware loss weights, with the x-axis reported in training iterations. Overly large weights slow convergence and reduce the final plateau, while lightweight regularization improves late-stage performance. The setting $(\lambda_e, \lambda_f) = (0.1, 0.02)$ achieves the best convergence trajectory and final DSC.

Qualitative Results. Fig. 5 shows that VM-UNet exhibits characteristic SHFA-induced boundary erosion, while FWMamba-UNet recovers sharp organ contours through parallel frequency compensation.

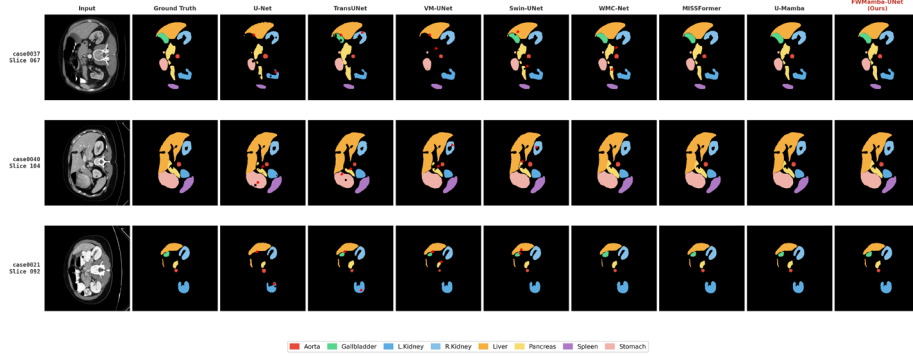


Fig. 5. Qualitative comparison on Synapse. FWMamba-UNet recovers fine organ contours (especially gallbladder and pancreas) that VM-UNet’s SHFA erodes.

5 Conclusion

We identified Structural High-Frequency Attenuation (SHFA) as an intrinsic limitation of HiPPO-initialized SSMs and proposed FWMamba-UNet to compensate for it through two modules: the FW-Mamba Block for parallel wavelet-based high-frequency enhancement, and EAFF-Skip for boundary-aware feature propagation. With only +10.1% parameters, the model achieves state-of-the-art performance on Synapse, ACDC, and ISIC 2018, and our ablations indicate that SHFA is a feature-representation deficiency that must be addressed architecturally.

Limitations and Future Work. Our study is scoped to 2D, fully-supervised, single-modality medical segmentation, leaving several broader settings open. Clinical practice is increasingly volumetric and multi-modal, so extending the wavelet pathway to anisotropic 3D bases and cross-modal fusion (e.g., CT+MRI) is a natural next step. Annotation cost also makes semi-supervised and few-shot regimes more realistic than full supervision, yet the interaction between SHFA and label scarcity remains unexplored. Finally, boundary-sensitive vision tasks beyond medicine—remote-sensing delineation, document layout parsing, video matting, and fine-grained temporal recognition—are likely to share the same bias, and validating our framework across these domains is a promising direction for future SSM-based vision models.

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